Zhang 2022 HGSOC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

20/22 samples (≥41 cells) / 11 patients intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached); both lobes (Body 8,002 / Terminal cap 756) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

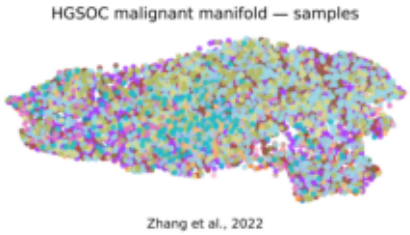
Gast/pSom co-vary (r=+0.55) and decline along pseudotime (ρ=−0.65/−0.80); Crest anti-correlated (r=−0.83/−0.82), rising to a terminal readout (ρ=+0.87; crest-high decile mean pt=0.94).

Gate 3 — hub mutual exclusivity → NA

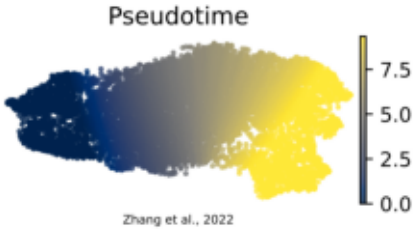
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

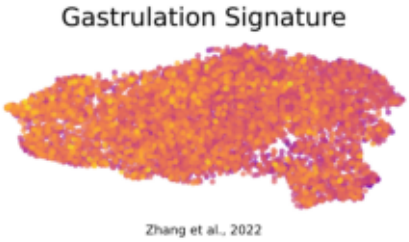
Manifold — samples



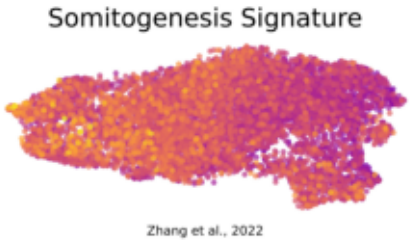
Pseudotime



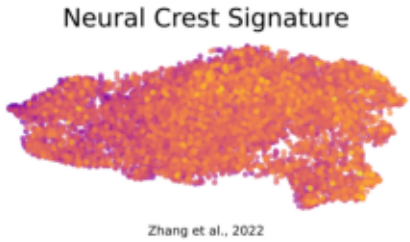
Gastrulation



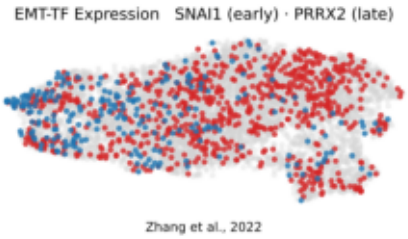
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

High-Grade Serous Ovarian Carcinoma (Zhang et al., 2022), 10x — 22 samples / 11 patients / 51,786 cells (master scope); malignant only: 8,806 load, 8,758 across 20 samples (≥41) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.

Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zhang's HGSOC malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The developmental axis is orthogonal to cycling (ρ=−0.003) — no proliferation confound, and both lobes are low-cycling, so the whole manifold is kept (Bischoff/Sun no-scoping precedent). EMT-TFs SNAI1 (early/root) → PRRX2 (late/terminus); canonical ZEB1/TWIST near the detection floor (6.5%/3.1%); EMT-TF end-separation weak (Δpt=0.29).

Parameter choices

Min cells/sample	≥41 (20/22; gap 57→30)	**Integration**	Harmony θ=5; PCA 50PC; spread=1.3
Scope	whole manifold (no lobe scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Compartments**	Body 8,002 / Terminal cap 756